MILITARY SPECIFICATION

FIBER, SHREDDED WOOD

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1.0 SCOPE

1.1 Scope. - This specification establishes the requirements and quality assurance provisions for the manufacture and acceptance of one grade of shredded wood fiber for use in inert components for combustible cartridge cases. (See 6.1.)

2.0 APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards and handbooks. - Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

BENEFICIAL COMMENTS (RECOMMENDATIONS, ADDITIONS, DELETIONS) AND ANY PERTINENT DATA WHICH MAY BE OF USE IN IMPROVING THIS DOCUMENT SHOULD BE ADDRESSED TO: COMMANDER, US ARMY ARMAMENT MUNITIONS AND CHEMICAL COMMAND, ATTN: DRSMC-QA, DOVER, NEW JERSEY 07801 BY USING THE SELF-ADDRESED STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL (DD FORM 1426) APPEARING AT THE END OF THIS DOCUMENT OR BY LETTER.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:28Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E300	Recommended Practice for Sampling Industrial Chemicals

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI) Official Test Methods

T-207 OM-81	Water Solubility of Wood

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:28Z
Check the source to verify that this is the current version before use.

T-210 M-58	Weighing, Sampling and Testing Pulp for Moisture
T-211	Ash in Wood and Pulp
UM-2	Density of Pulpwood
UM-21	Sieve Analysis of Pulpwood
UM-203	Freeness of Pulp (Williams Tester)

(Application for copies should be made to the Technical Association of the Pulp and Paper Industry, Technology Park, P.O. Box 105113, Atlanta, GA, 30348.)

2.3 Order of precedence. - In the event of conflict between the text of this specification and the referenced specifications cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. - The shredded wood fiber shall comply with requirements given in this detailed specification as well as applicable drawings and referenced specifications.

3.2 Material. - The material shall be shredded wood fiber, meeting the requirements of this specification.

3.3 Properties. - The properties of the shredded wood shall be as specified in Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:28Z
Check the source to verify that this is the current version before use.

Table I

Property	Requirement	Applicable Test Method
Moisture Content	15% Maximum	4.5.1
Ash Content	0.8% Maximum	4.5.2
Particle Size		4.5.3
Fines	8.0% Maximum	
Oversized Chips	3.0% Maximum	
Williams Freeness	6-7 seconds	4.5.4
Density	0.25-0.35 g/cc	4.5.5
Water Soluble Content		4.5.6
Hot Water	6-9%	
Cold Water	4-6%	

3.4 First article sample. - This specification makes provision for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. - The best commercial practices shall be used in the manufacture of the product furnished under this specification. Procedures and controls shall be employed which preclude conditions which may adversely affect the safety and reliability of the product (i.e., contaminated vessels, impurities, grit, chips, etc.), when tested in accordance with 4.5.7.

4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such specifications are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:28Z
Check the source to verify that this is the current version before use.

specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

4.2 Classification of inspections. - The inspection requirements specified herein are classified as follows:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall be 1 kilogram of shredded wood fiber. The sample shall be obtained from the first production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspection to be performed. - The sample will be subjected by the Government to any or all of the examinations or tests specified in Table II.

4.3.3 Rejection. - See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Lot formation. - A lot shall consist of one or more batches of shredded wood fiber produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quality of shredded wood fiber that has been subjected to the same chemical or physical process intended to make the final product homogeneous.

4.4.2 Examination. - See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:28Z
Check the source to verify that this is the current version before use.

TABLE II. FIRST ARTICLE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Fiber, Shredded Wood				N/A
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Moisture Content			3.3	4.5.1/Lab analysis
	Ash Content			3.3	4.5.2/Lab analysis
	Particle Size			3.3	4.5.3/Lab analysis
	Williams Freeness			3.3	4.5.4/Lab analysis
	Density			3.3	4.5.5/Lab analysis
	Water Soluble Content			3.3	4.5.6/Lab analysis
	Workmanship			3.5	4.5.7/Visual

NOTES:

HC-685

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:28Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS AND TESTS

DOD-F-70459 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Fiber, Shredded Wood				N/A
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Moisture Content	1/		3.3	4.5.1/Lab analysis
102	Ash Content	1/		3.3	4.5.2/Lab analysis
103	Particle Size	1/		3.3	4.5.3/Lab analysis
104	Williams Freeness	1/		3.3	4.5.4/Lab analysis
105	Density	1/		3.3	4.5.5/Lab analysis
106	Water Soluble Content	1/		3.3	4.5.6/Lab analysis
Minor					
201	Workmanship	1/		3.3	4.5.7/Visual

NOTES:
1/ See paragraph 4.4.3.1

HC-685

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:28Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. - It is emphasized that all applicable safety rules, regulations, practices and procedures be followed in handling and processing the shredded wood fiber.

4.4.3.1 Sampling. - A representative sample shall be withdrawn from each batch or lot of shredded wood fiber, in accordance with paragraph 28.4 of ASTM E300. Each sample shall be tested independently in accordance with the applicable methods given in 4.5. If the sample fails to meet any of the test requirements, the batch or lot shall be subjected to a retest. If, upon retesting, a subsequent sample fails to meet any of the test requirements, the batch or lot shall be rejected. MIL-STD-105 shall apply for the defect classification given in paragraph 4.4.2.1.

4.4.4 Inspection equipment. - For the performance of all tests and examinations specified in 4.4 and 4.5, commercial inspection and laboratory equipment shall be employed. The contractor shall have available, and utilize correctly, this equipment and is charged with the responsibility of assuring that proper calibrating procedures are followed.

4.5 Test methods and procedures. - All tests given in this section shall be performed using prescribed procedures given in the Technical Association of the Pulp and Paper Industry (TAPPI) Official Test Methods. See 6.3 for use of equivalent test methods.

4.5.1 Moisture content. - The sample shall be tested for moisture content in accordance with TAPPI Test Method T-210 M-58.

4.5.2 Ash content. - The sample shall be tested for ash content in accordance with TAPPI Test Method T-211.

4.5.3 Particle size. - The sample shall be tested for particle size in accordance with TAPPI Method UM-21 with the following modifications:

a. The screen sizes (mesh) used shall be: 5 and 100.

b. All fiber bundles are manually separated before sampling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:28Z
Check the source to verify that this is the current version before use.

c. After 5 minutes of shaking time, the sieve shaker is turned off and the sample is visually examined for any remaining fiber bundles in the top screen. If any bundles are present, they are manually separated and shaking is continued.

d. Operation c is repeated as necessary.

e. The total shaking time should not exceed 45 minutes.

4.5.3.1 The oversized chip percentage is defined as that portion retained on the 5 mesh screen and the fines percentage is that portion passing the 100 mesh screen. The percent oversized chips and percent fines are calculated using the following equations:

$$\% \text{ oversize chips} = \frac{\text{wt of particles retained on 5 mesh screen}}{\text{original wt of sample}} \times 100$$

$$\% \text{ fines} = \frac{\text{wt of particles retained on 100 mesh screen}}{\text{original wt of sample}} \times 100$$

4.5.4 Williams freeness. - The samples shall be tested for Williams Freeness in accordance with TAPPI Test Method UM-203.

4.5.5 Density. - The sample shall be tested for density in accordance with TAPPI Test Method UM-2.

4.5.6 Water Soluble Content. - The sample shall be tested for hot and cold water solubility in accordance with TAPPI Test Method T-207 OM-81.

4.5.7 Workmanship. - The sample, when visually examined, shall be free from foreign matter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:28Z
Check the source to verify that this is the current version before use.

5.0 PACKAGING

5.1 Packing

5.1.1 Level C. - The wood fiber shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

5.2 Marking. - Unless otherwise specified, marking shall be in accordance with MIL-STD-129.

6.0 NOTES

6.1 Intended use. - Shredded wood fiber is intended to be used in inert components of 120mm combustible cartridge cases.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

6.3 Equivalent test methods. - The test methods given in this specification are the official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (accuracy and precision) to the method given in this specification. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Technical Agency to: COMMANDER, ARDC, ATTN: DRSMC-QAR-R, Dover, NJ 07801. This description should include, but not be limited to, the procedures used, the accuracy and precision and drawings of any special equipment required (see MIL-I-45208, paragraph 3.10).

Preparing Activity
Army-AR
Project Number: 1320-A784

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:28Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-F-70459	FIBER, SHREDDED WOOD

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:28Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDER

US ARMY ARMAMENT MINITIONS & CHEMICAL COMMAND
ATTN: DRSMC-QA
DOVER, NJ 07801 5001

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:28Z
Check the source to verify that this is the current version before use.